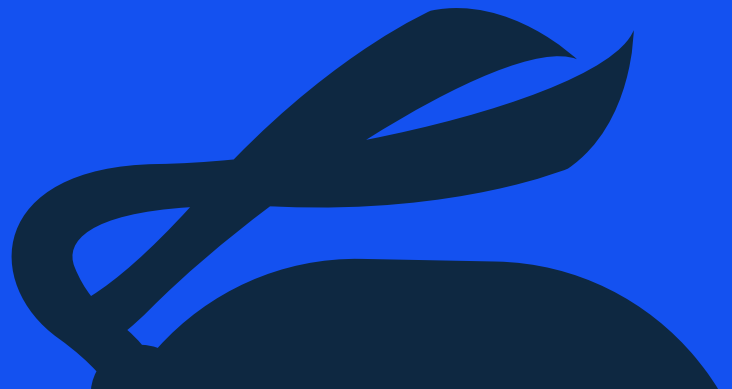
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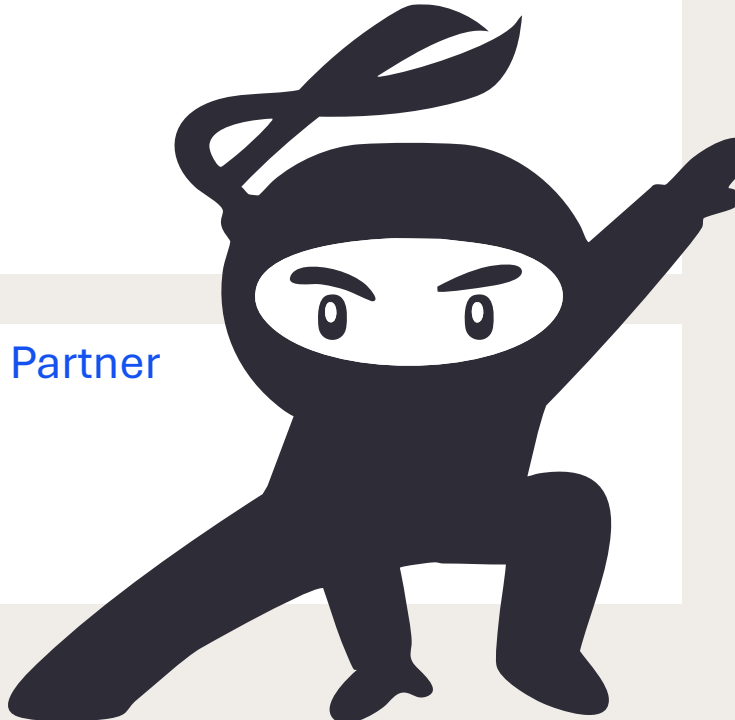
Bronze

ugurlabs

iC CONSULT

Snacks & Networking Partner

SCE



Defender Boys



Speaker

Mattias Borg

DFIR, Onevinn AB

Microsoft MVP Most Valuable Professional



Speaker

Stefan Schörling

DFIR, Onevinn AB

Microsoft MVP Most Valuable Professional

Session

Hidden gems in and around Defender XDR with Defender Boys

Some unknowns, some reminders, some underground



#SecNinjas

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Deployment & Onboarding



MDE - Onboarding

- Review All Exclusions, Start-over with a blank paper
- Make sure you do a Full-scan after onboarding
- Decide on Data Sharing Settings
- *Cloud Protection, Sample Submission, Cloud Backup, AIR, Analyst Actions like File Collection*
- Decide on your Management Model
- *Who does what, Protect, Detect, Govern, Remediation*
- *Not all settings are supported on Defender Settings Management yet*

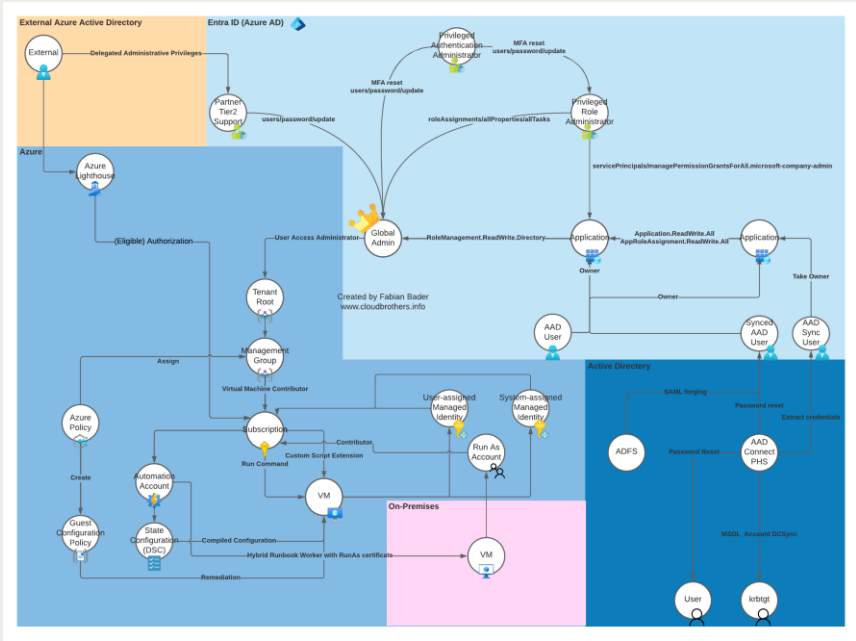
MDE - Selective Response Actions

Control high-impact response actions on Tier-0 systems and other high-value assets.

Protection against bad behavior

Information Collection
- Collect Files

Lateral Movement & Disruption
- Enable Live Response
- Change Device Groups
- Run Live Response (API or Console)



Source: https://cloudbrothers.info/azure-attack-paths/

MDE - Selective Response Actions

Plan, Do, Act



Configured during onboarding

Enable the feature, generate an onboarding package with restricted security operations, then onboard the device and verify its status.

Plan restrictions carefully: changing them requires offboarding and onboarding again with a new package.

Home - Microsoft Defender

https://security.microsoft.com/homepage?tid=60b7592b-e432-4dbc-b9b5-d2f10cbede92

Microsoft Defender

Search

Home

Incidents

Advanced hunting

Hide navigation

Find Defender solutions

Home

Cases New

Exposure management

Investigation & response

Threat intelligence

Assets

Microsoft Sentinel

Identities

Endpoints

Email & collaboration

Cloud apps

Cloud security

SOC optimization

Reports

Learning hub

Trials

More resources

Setup & configuration

Audit

Data connectors

Pay as you go

Permissions

Health

Settings

Home

Light mod

You have devices running Defender for Endpoint via the Microsoft Monitoring Agent (MMA), which will stop working on February 26, 2027. Prevent gaps in protection by upgrading your devices to the latest agent before the deadline.

Your unified SIEM and XDR is ready

Everything you need is now in one place. You can now hunt, investigate, and respond faster with full attack context across both SIEM and XDR.

Start hunting

Learn about unified SIEM and XDR experiences

Guided tour

What's new?

Community

Add c

Entity Behavior Analytics (UEBA)

Workspace: log-securitydata

No anomalies found in the last 14 days

Data sources

Data sources with UEBA enabled

0/12

View UEBA Settings

SOC optimization

Your optimization data

Optimization status

Active 14

In progress 0

Completed 37

Dismissed 0

Recent optimization value

You improved your coverage against X-Cloud Attacks attacks.

Utilized SigninLogs table and improved your SOC coverage.

Microsoft Sentinel data connectors

2 data connectors

Last 24 hours

Active connectors 2/2

Unhealthy connectors 0/2

TI by type (0)

Data received

9:00 13:00 17:00 21:00 1:00 5:00

1/2 Events (367) Events in previous 24 hours (586)

Identity Security Deployment Status

Defender protects your identities with posture security recommendations and threat protection. Learn more

No data available

Improve your identity protection

Connected SaaS apps

Defender for Cloud Apps protects your SaaS apps wit security configuration recommendations, threat prote and information protection. Learn more

SaaS app connectors by health status

Healthy 1

Needs attention 0

Connection errors 0

Connect your SaaS apps

Microsoft Secure Score

Secure Score: 53.5%

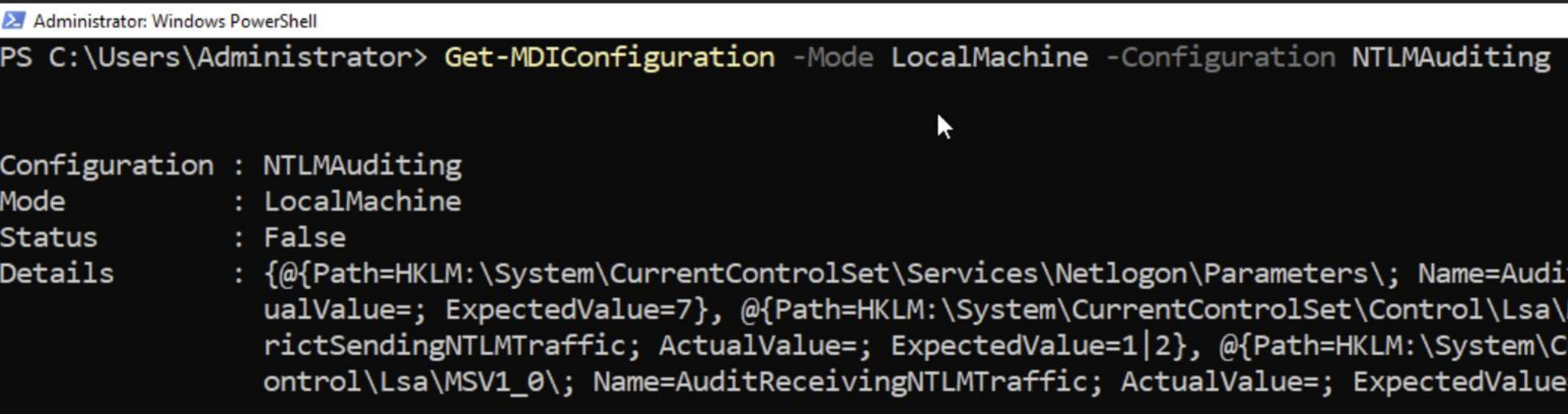
719.09/1344 points achieved

Microsoft Secure Score is a representation of your organization's security posture, and your opportunity improve it.

Score last calculated 10/17

MDI - Readiness

Clear-MDISensorProxyConfiguration	Clears the proxy configuration for the Defender for Identity sensor.
Get-MDIConfiguration	Gets the configuration for various Defender for Identity post-deployment required settings.
Get-MDISensorProxyConfiguration	Retrieves the proxy configuration for the Defender for Identity sensor.
New-MDIConfigurationReport	Generates a report of MDI domain configuration and saves it as an HTML and a detailed JSON file.
New-MDIDSA	Creates a new Directory Service Account (DSA).
Set-MDIConfiguration	Sets the configuration for various Defender for Identity post-deployment required settings.
Set-MDISensorProxyConfiguration	Sets the proxy configuration for the Defender for Identity sensor.
Test-MDIConfiguration	Validates the configuration for various Defender for Identity post-deployment required settings.
Test-MDIDSA	Validates the permissions and delegation of a Directory Service Account (DSA).
Test-MDISensorApiConnection	Validates the connection to the Defender for Identity Sensor API.

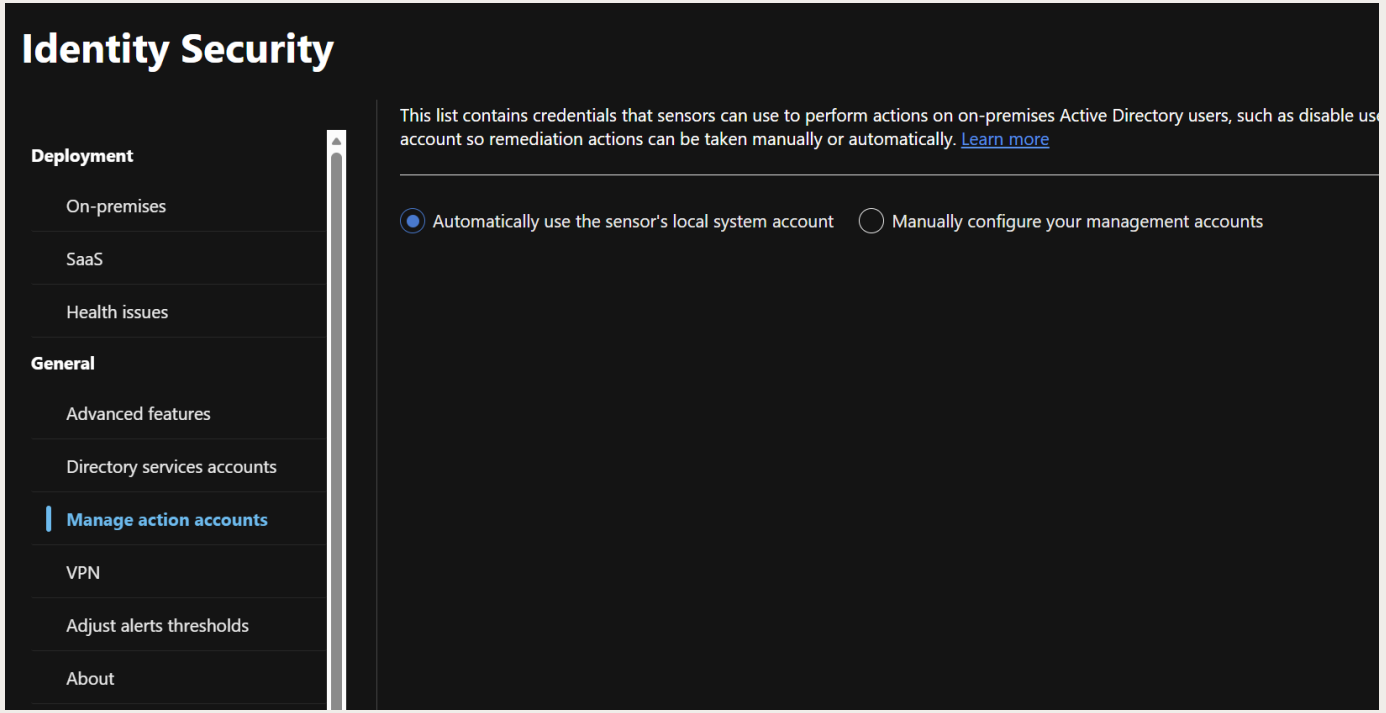


MDI - v3 Sensors

Change Managed Service Account on Sensors to Local system

Remove unused gMSA Accounts

Remove unused NPCAP

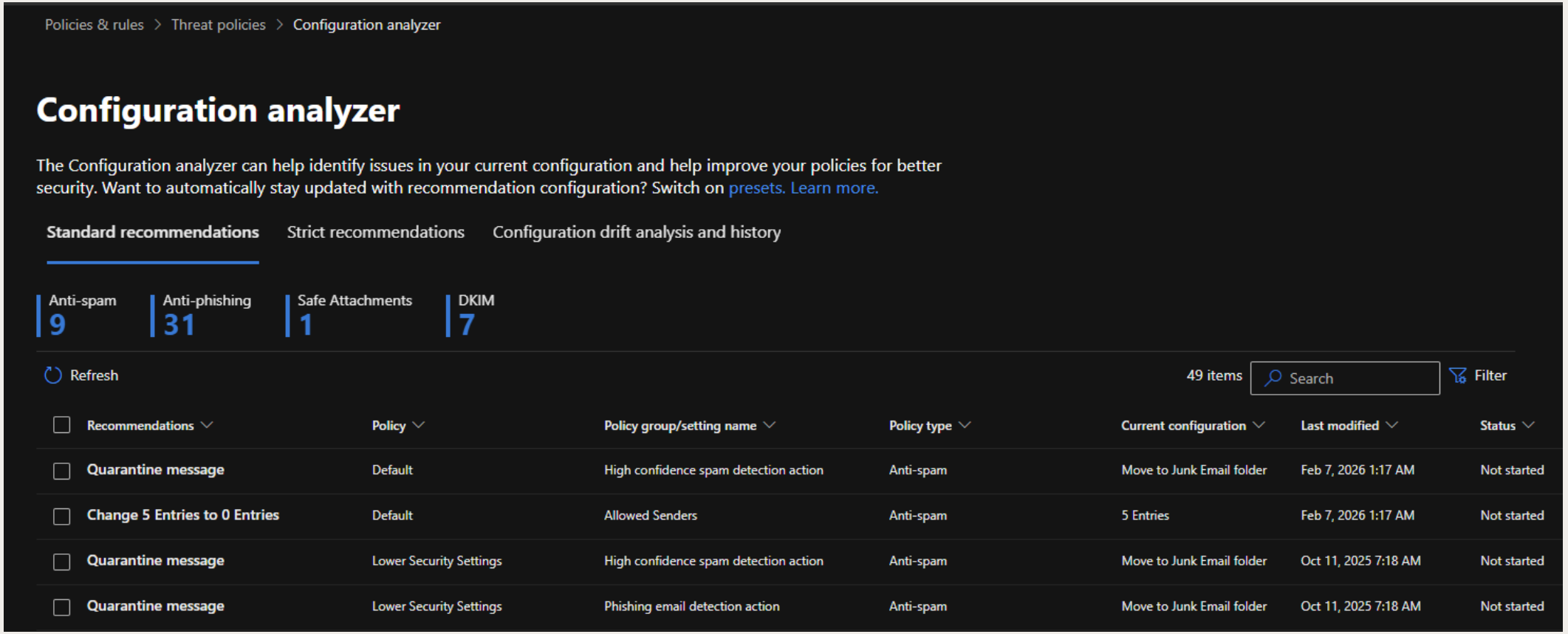


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Configuration



MDO – Configuration Analyzer



MDE - Contextual Exclusions

What if I could limit my exclusions even further

Contextual restriction type	Keyword	Value
File and folder restriction	`PathType`	`file` `folder`
Scan type restriction	`ScanType`	`quick` `full`

Scan trigger restriction	`ScanTrigger`	`OnDemand`
		`OnAccess`
		`BM` (Behavior monitoring)

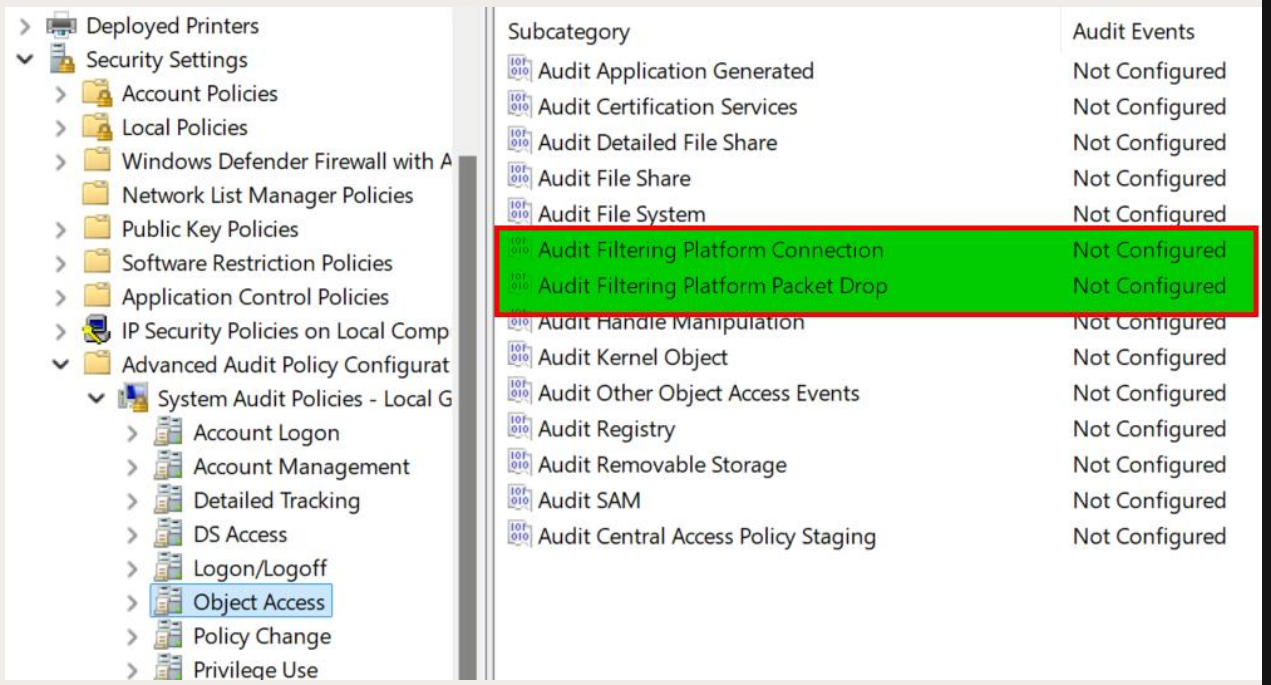
Process restriction	`Process`	`<path>`

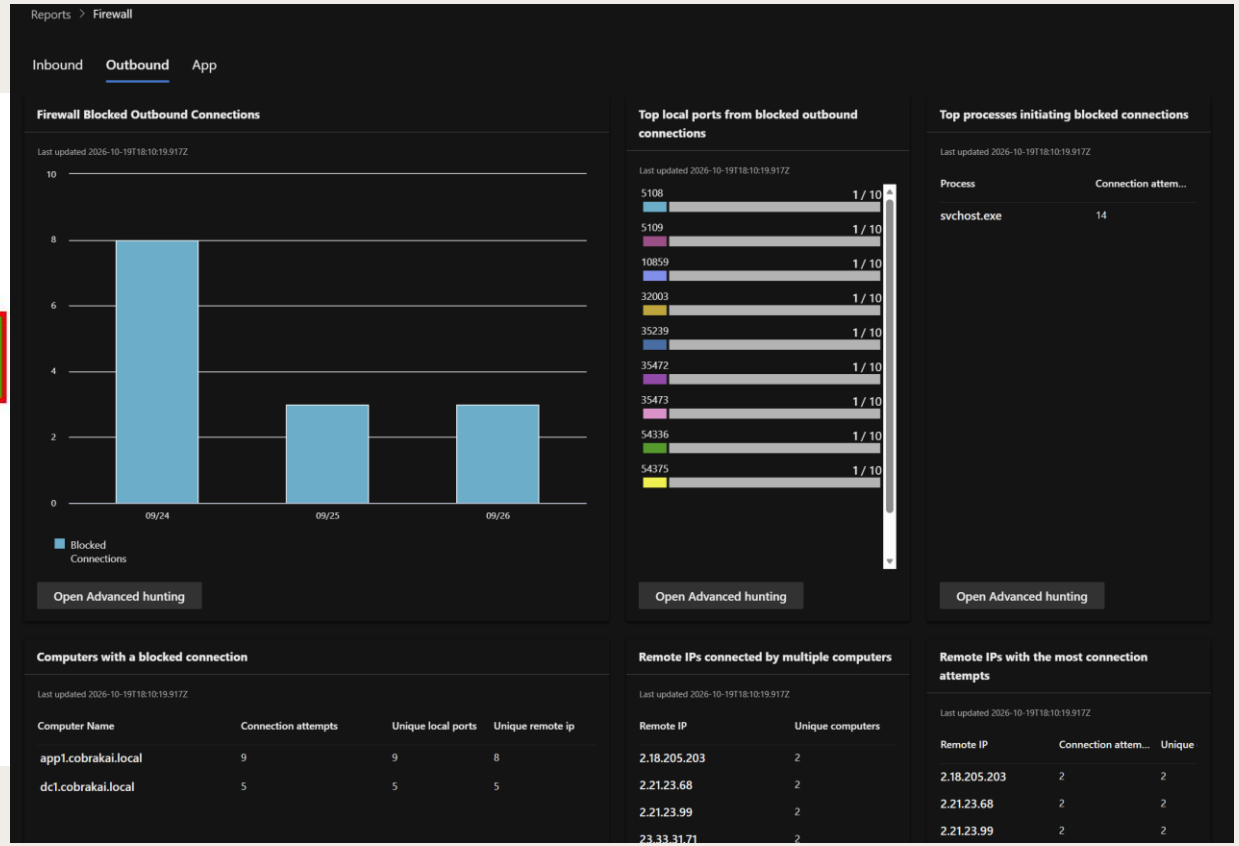
```
c:\documents\design.docx\:{PathType:file,ScanTrigger:OnDemand}
```

MDE - Windows Firewall Audit

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FW Audit != FW Logging





https://learn.microsoft.com/en-us/defender-endpoint/host-firewall-reporting

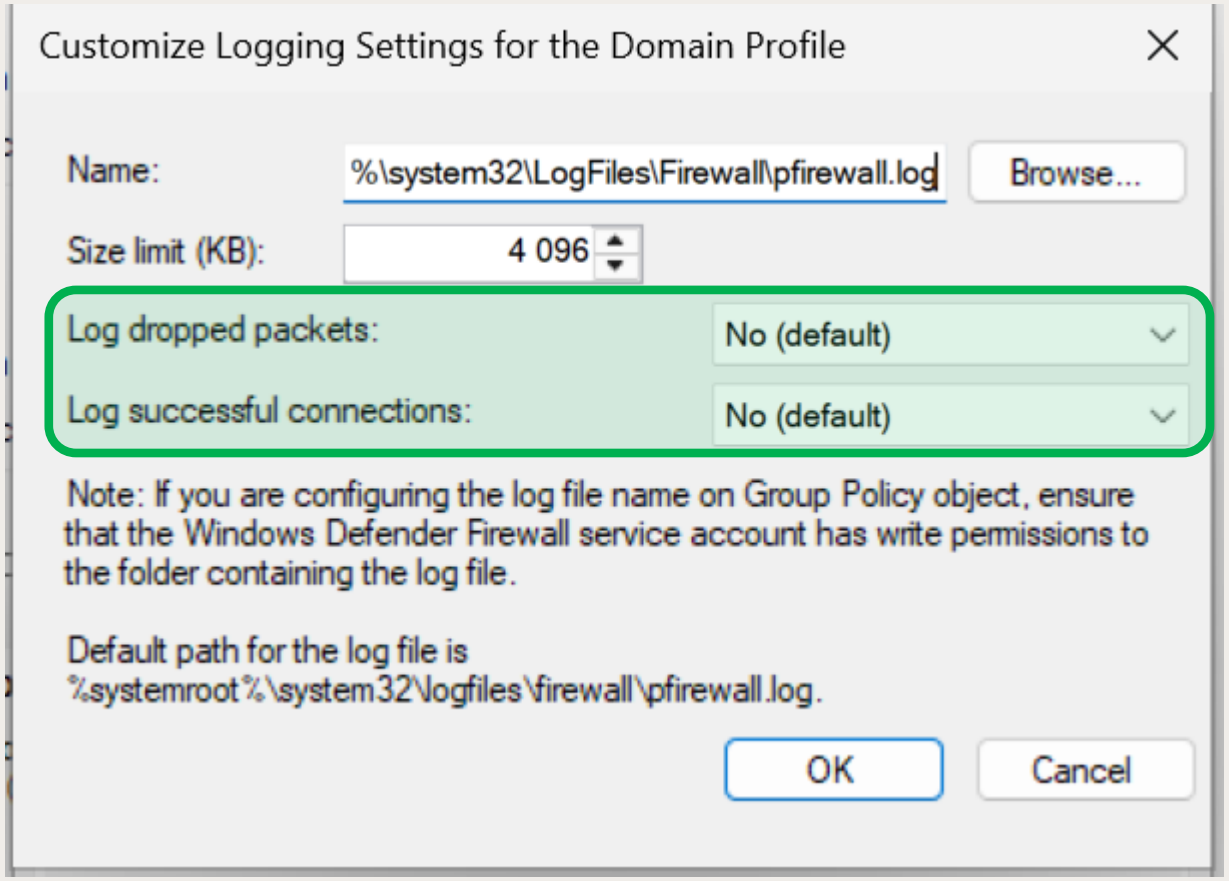
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MDE - Windows Firewall Log

Let’s put things in that log

Keep name: **pfirewall.log and path)**
(if you want it in the investigation package)



MDE - Antimalware Default actions



ALWAYS QUARANTINE

Even on LOW

Alerts multiple tenants 7d

Low	13
Low	47
Low	54
Low	14
Low	4
Low	1
Low	18
Low	95

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MDE - Sample Submissions

Send safe samples automatically

- Safe samples are samples considered to not commonly contain PII data. Examples include .bat, .scr, .dll, and .exe.
- If file is likely to contain PII, the user gets a request to allow file sample submission.
- This option is the default configuration on Windows, macOS, and Linux.

Always Prompt

- If configured, the user is always prompted for consent before file submission
- This setting isn't available in macOS and Linux cloud protection

Send all samples automatically

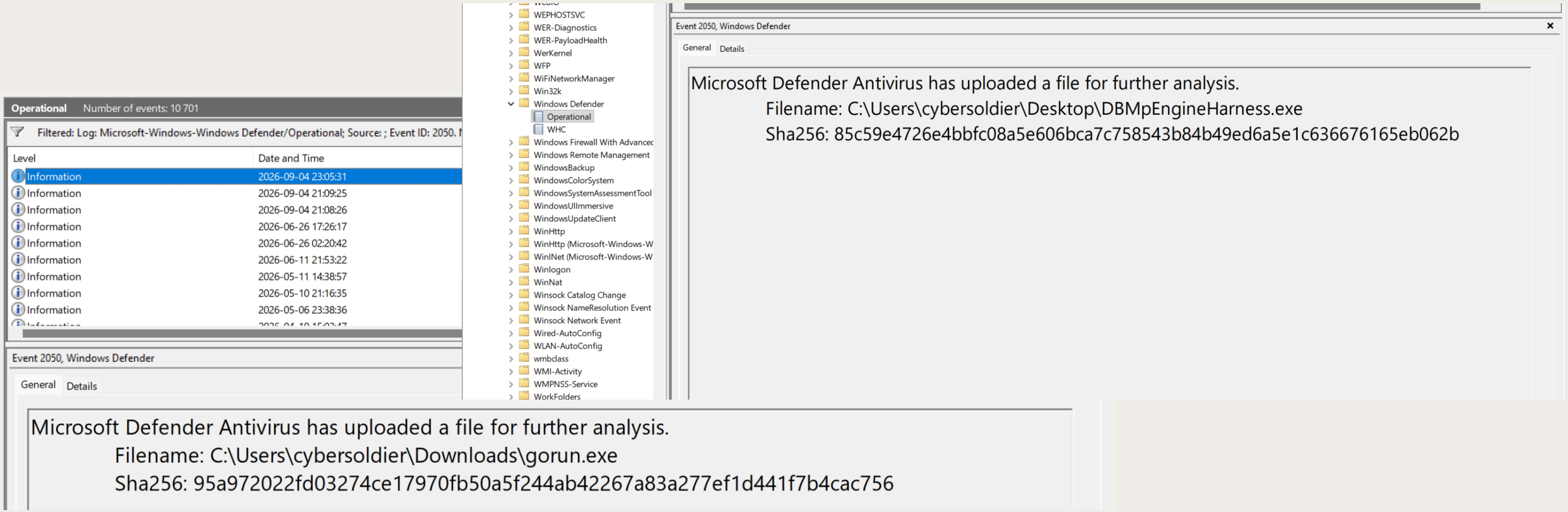
- If configured, all samples are sent automatically
- If you would like sample submission to include macros embedded in Word docs, you must choose **Send all samples automatically**
- This setting isn't available on macOS cloud protection

Do not send

- Prevents "block at first sight" based on file sample analysis
- "Don't send" is the equivalent to the "Disabled" setting in macOS policy and "None" setting in Linux policy.
- Metadata is sent for detections even when sample submission is disabled

MDE – Sample Submission

What files are really being sent to Microsoft?



MDE - Sample Submissions

What files are really being sent to Microsoft?

Event log >>

Advanced Hunting

```
union DeviceNetworkEvents, DeviceProcessEvents
| where (FileName == "MpCmdRun.exe"
        and ProcessCommandLine has "SampleHeaderService"
        )
        or InitiatingProcessCommandLine has "SampleHeaderService"
```

Event ID 2050

Symbolic name: `MALWAREPROTECTION_SAMPLESUBMISSION_UPLOADED`

Message: The antimalware platform submitted a file for sample analysis.

Description: A file was submitted to the cloud for further analysis as part of the automatic sample submission feature.

- File Name: The full path of the file that was submitted for analysis.
- SHA-256 Hash: The SHA-256 hash of the submitted file.

User action: No action is required. This event is informational. The file was submitted based on the sample submission consent configuration. By default, Microsoft Defender Antivirus automatically sends files that don't contain personally identifiable information (PII). For more information about sample submission settings, see Cloud protection and sample submission at Microsoft Defender Antivirus.

MDE - Aggregated Reporting

Defender for Endpoint P2 summarizes supported endpoint events over one-hour intervals while retaining essential properties for investigation and hunting.

This adds visibility into repetitive, low-signal activity that normal reporting can cap. Retaining more telemetry can increase downstream SIEM or storage costs.

Supported tables

- DeviceFileEvents
- DeviceNetworkEvents
- DeviceProcessEvents
- DeviceLogonEvents

Additional visibility into repetitive file activity, network beaconing, etc

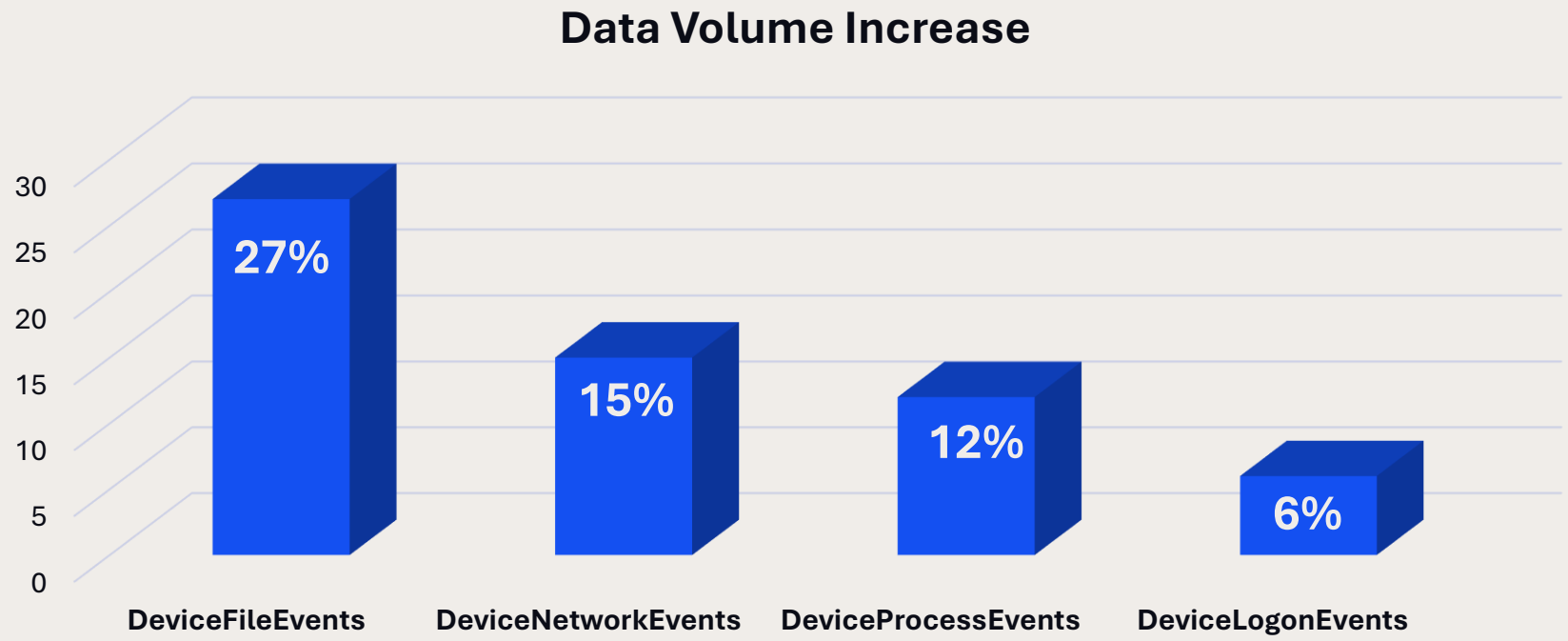
Enable: Settings > Endpoints > Advanced features > Aggregated reporting

MDE - Aggregated Reporting

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How much will the Data Volume increase?

(will differ per tenant, data is a average across a subset of tenants)



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MDE – Local Agent Protection

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Platform

Windows

Set-AINetworkInspection.ps1 > ...

```powershell
try {
    Set-MpPreference -AiAgentNetworkInspection Audit -ErrorAction Stop
    $isAudit = & (Join-Path $PSScriptRoot 'Get-AINetworkInspection.ps1')

    if ($isAudit -eq $true) {
        exit 0
    }

    Write-Error 'AI agent network inspection was not set to Audit.' -ErrorAction Continue
    exit 1
}
catch {
    Write-Error -ErrorRecord $_ -ErrorAction Continue
    exit 1
}
```

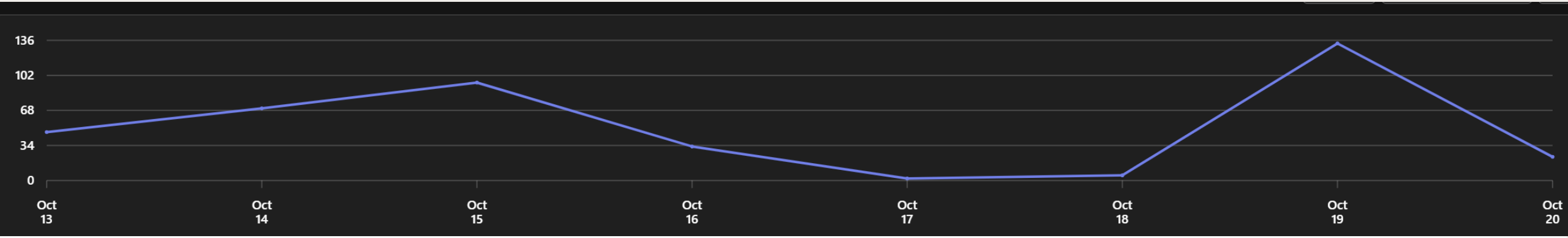
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MDO - Tuning

Quarantined but released by end users

```
1    EmailPostDeliveryEvents
2    | where Action == "Quarantine release"
```

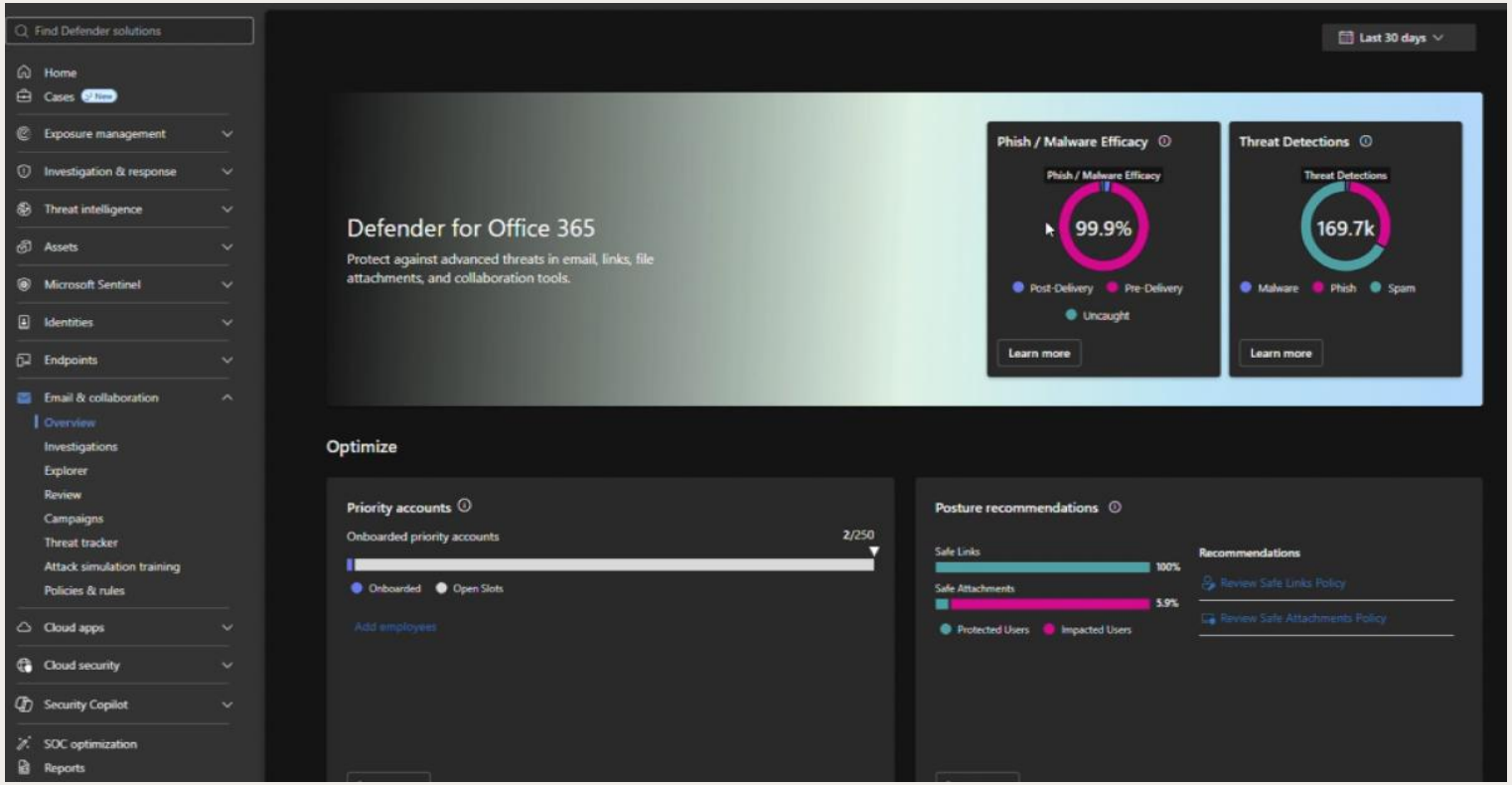


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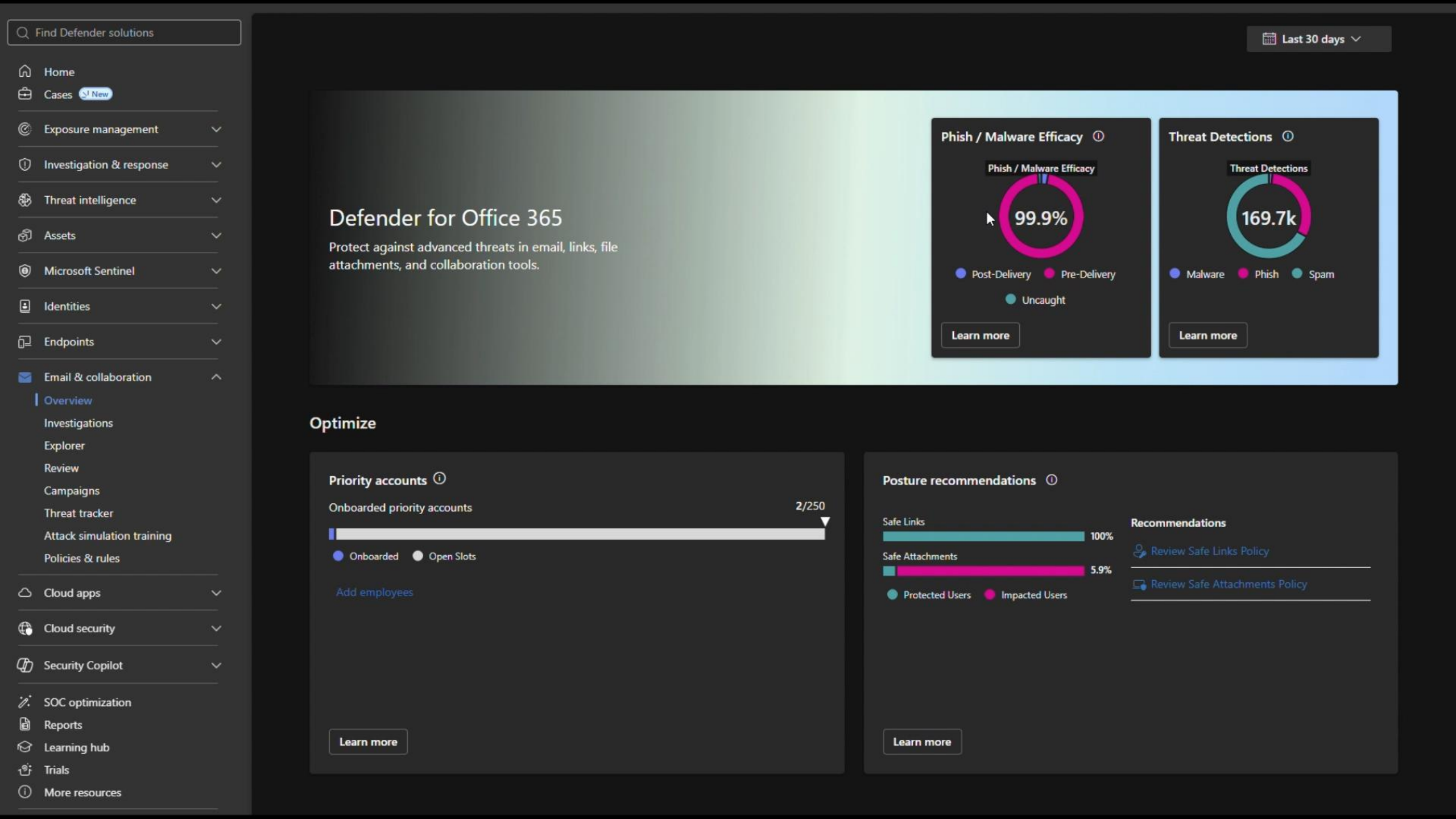
MDO – Mailflow Review

Overview Page



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Find Defender solutions
Home
Cases
New
Exposure management
Investigation & response
Threat intelligence
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Microsoft Sentinel
Identities
Endpoints
Email & collaboration
Overview
Investigations
Explorer
Review
Campaigns
Threat tracker
Attack simulation training
Policies & rules
Cloud apps
Cloud security
Security Copilot
SOC optimization
Reports
Learning hub
Trials
More resources
Last 30 days
Defender for Office 365
Protect against advanced threats in email, links, file attachments, and collaboration tools.
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99.9%
Post-Delivery
Pre-Delivery
Uncaught
Learn more
Threat Detections
Threat Detections
169.7k
Malware
Phish
Spam
Learn more
Optimize
Priority accounts
Onboarded priority accounts
2/250
Onboarded
Open Slots
Add employees
Learn more
Posture recommendations
Safe Links
100%
Safe Attachments
5.9%
Protected Users
Impacted Users
Recommendations
Review Safe Links Policy
Review Safe Attachments Policy
Learn more

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Security Operations



MDE - Live Response

Background jobs



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Live response on dc01

Connected

Disconnect session

Command console | Command log

Maxi

Command index

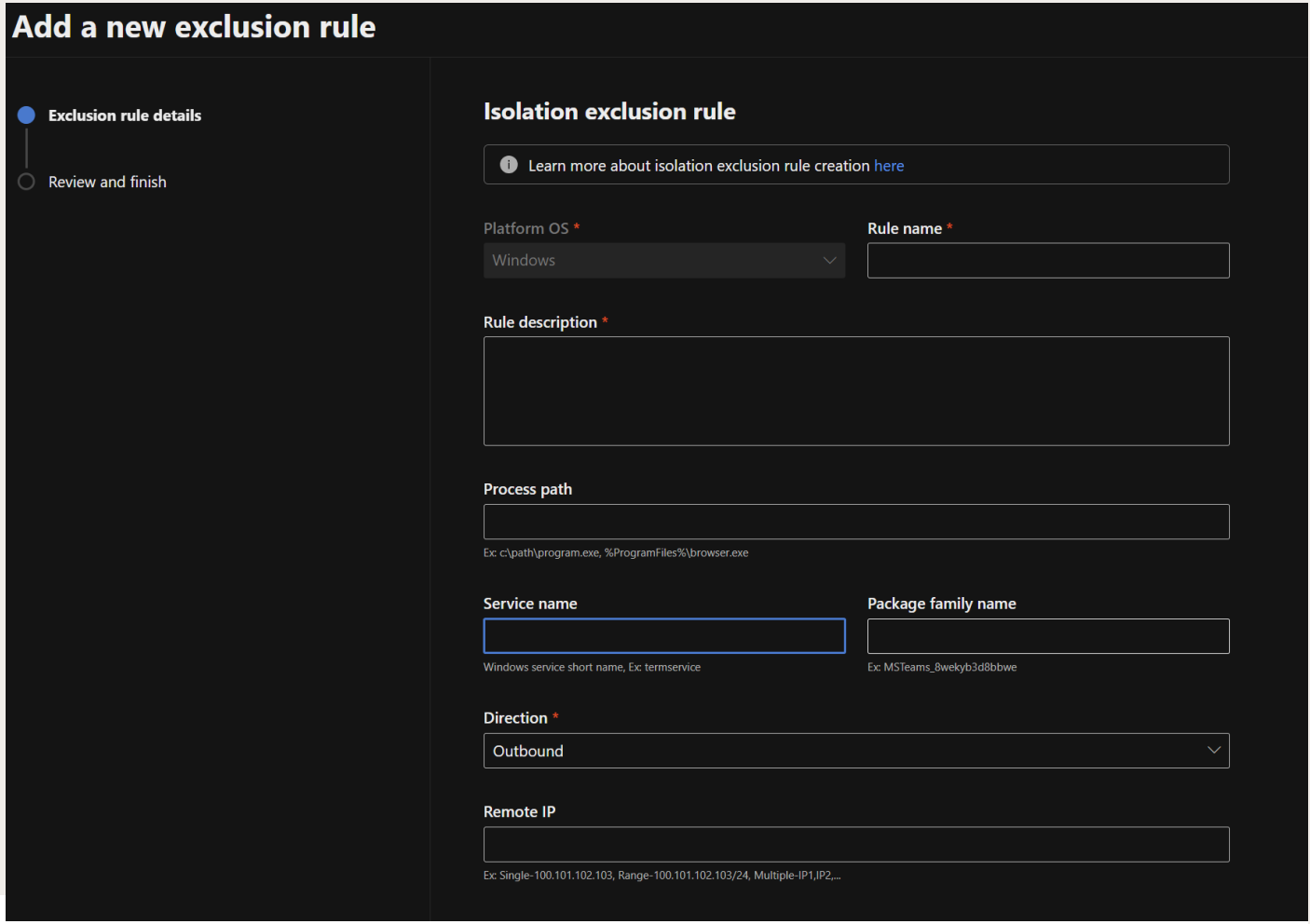
```
C:\> clear

C:\> run Invoke-ADForensics.ps1
```

MDE - Isolation exclusions

Exclude In/Out

Use for your evidence collection platform



MDE - DisruptionAndResponseEvents

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Advanced hunting

New query* × | New query* × | New query* × | New query* × | New query* ×

Query | Run query | All tenants, All

```
1   DisruptionAndResponseEvents
2   | where ActionType == 'ContainedRestrictedUserSmbFileOpenBlocked' or
3           ActionType == 'ContainedRestrictedUserSmbFileOpenBlocked' or
4           ActionType == 'ContainedUserLogonBlocked' or
5           ActionType == 'ContainedUserLogonBlockedByDomainController' or
6           ActionType == 'ContainedUserRemoteDesktopSessionDisconnected' or
7           ActionType == 'ContainedUserRemoteDesktopSessionStopped' or
8           ActionType == 'ContainedUserRpcAccessBlocked' or
9           ActionType == 'ContainedUserSmbFileOpenBlocked' or
10          ActionType == 'ContainedUserSmbFileOpenBlockedAggregation' or
11          ActionType == 'ContainedUserSmbSessionStopped' or
12          ActionType == 'GroupPolicyAccessBlocked' or
13          ActionType == 'GroupPolicyHardeningPolicyApplied' or
14          ActionType == 'GroupPolicyHardeningPolicyRemoved' or
15          ActionType == 'SafeBootBlocked' or
16          ActionType == 'SafeBootGuardPolicyApplied' or
17          ActionType == 'SafeBootGuardPolicyRemoved'
```

```
1   DisruptionAndResponseEvents
2   | summarize arg_max(TimeGenerated,*) by ActionType
```

Results | Getting started

Export | Show empty columns

Filters: Add filter

TimeGenerated	ActionType
Oct 19, 2026 12:57...	ContainedUserLogonBlocked
Oct 19, 2026 12:47...	ContainedUserSmbFileOpenBlocked
Oct 19, 2026 12:47...	ContainedUserRemoteDesktopSessionDisconnected
Oct 19, 2026 12:47...	ContainedUserRemoteDesktopSessionStopped
Oct 19, 2026 9:42:...	ManuallyContainedDeviceBlocked
Oct 19, 2026 2:00:...	ContainedUserSmbFileOpenBlocked

https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-disruptionandresponseevents-table

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XDR - Hunting Functions

AssignedIPAddresses()

DeviceFromIP()

FileProfile()

SeenBy()

```
let userIsDomainAdmin = (T:(Sid:string))
{
    let DomainAdminSids =
        ExposureGraphNodes
        | where NodeLabel == "user"
        | extend json = parse_json(NodeProperties).rawData
        | where json.nestedAdGroupNames has "Domain Admins"
        | project Sid = tostring(json.adSid)
        | where isnotempty(Sid)
        | distinct Sid;
    T
    | lookup kind=leftouter (DomainAdminSids | extend IsDomainAdmin = true) on Sid
    | extend IsDomainAdmin = coalesce(IsDomainAdmin, false)
};
DeviceLogonEvents
| extend Sid = AccountSid
| invoke userIsDomainAdmin()
```

... and Custom Functions (User-Defined Functions)



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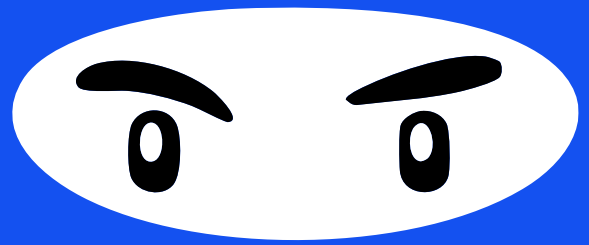
Defender Inside Out

Some research

Thank You, Ninjas!



See you next Year!



Ninjas

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